

Mohankumar Markuli Chandrayigowda

Portfolio @mohankumar-markuli in /in/mohankumar-mc

Email: mohankumarmarkuli@gmail.com

Mobile: +91 9187338626

PROFESSIONAL SUMMARY

A software engineer based in India, specializing in web technologies and AI-powered chatbot systems, Combining strong analytical thinking with expertise in backend engineering and scalable system design, the focus is on building high-performance architectures, seamless API integrations, and continuously exploring emerging technologies.

EDUCATION

University of Liverpool

Master of Science in Business Analytics and Big Data; CGPA: 6.7

Liverpool, United Kingdom

Sep 2022 – Dec 2023

Rajeev Institute of Technology

Bachelor of Engineering in Computer Science; CGPA: 7.41

Hassan, India

Aug 2016 – Sep 2020

EXPERIENCE

Infosys Limited

Systems Engineer

Bengaluru, India

Dec 2020 - Jul 2022

- Redesigned static web application to fully **responsive design using Angular and Bootstrap** MEAN stack product, expanding mobile accessibility for 500+ trainees during 3 month onboarding cycle for **GenQ assessment platform**.
- Analyzed learner performance data using **pandas, and numPy** across 23K to 27K daily active users, identifying top performers and reducing talent acquisition team screening time by 80% through automated candidate filtering for **Springboard EdTech Product**.
- Supported infrastructure for **Zoe chatbot microservice** integrated within Springboard, ensuring 0.999 uptime across enterprise services.
- Delivered **200+ plus training and sales** sessions on **AI, machine learning, big data, and Spark** to senior executives, team leads and on-boarding engineers, accelerating team ramp-up.

Infosys Limited

Software Engineering Internship – Spring Boot

Mysore, India

Feb 2020 – Mar 2020

- Built **REST APIs in Java and Spring Boot** handling 72+ product SKUs and orders, with **MySQL** queries optimized to sub-second retrieval times for Wintership ecommerce project.
- Structured layered architecture using **controller-service-repository** pattern, reducing code duplication by forty percent and enabling seamless **API testing** across 18 endpoints via **Postman**.

Adamos Technologies

Internship – Web Technologies and IOT

Mysore, India

Jul 2019 – Aug 2019

- Completed hands-on **web development** and **IoT** training covering 8 embedded systems modules across **Arduino** and **Raspberry Pi**, mastering microcontroller programming through practical exercises that directly enabled capstone project delivery.
- Designed and built proof-of-concept **advanced traffic management system** using **Raspberry Pi, IR sensors, and Python** for a 3 way and 4 way junction model, dynamically adjusting traffic signal timing based on vehicle density detection with real-time **data logging to Firebase cloud**.

PROJECTS

- PDF Loader RAG Pipeline** — Python, LangChain, OpenAI API or Ollama, Built an end-to-end Retrieval-Augmented Generation pipeline that loads PDF documents, chunks and embeds content into a vector store, and retrieves semantically relevant context to ground LLM responses, reducing hallucinations and enabling accurate document-based question answering.

- **Ethical Issues with the use of AI in Business Operations** — MSc Dissertation, University of Liverpool, December 2023. Research-driven analysis addressing three critical ethical dimensions: job displacement from automation, accountability when AI systems fail, and jurisdictional conflicts across different country regulations. Published on Medium.
- **News Aggregator API** — Node.js, Express, MongoDB, JWT, GNews API. Built personalized news aggregation backend with JWT authentication, user preference management, in-memory caching with background refresh to reduce external API calls, and full unit and integration test coverage.
- **FinEdge** Personal Finance & Expense Tracker API — Node.js, Express, MongoDB, JWT. Developed personal finance API with transaction CRUD, advanced filtering across five parameters, and MongoDB aggregation pipelines delivering monthly and yearly analytics with pagination support.
- **devTinder** Developer Networking Platform Backend — Node.js, Express, MongoDB. Built backend for a developer matchmaking platform featuring JWT authentication, profile management, and bi-directional connection request system with four interaction states enabling dynamic user feed generation.

CERTIFICATIONS

- **Airtribe AI first Software Engineer Program** (in Progress), Comprehensive backend engineering curriculum covering JavaScript fundamentals, REST APIs, asynchronous programming, microservices, JWT authentication, unit and integration testing, CI/CD with GitHub Actions, and emerging modules in RAG systems, LLM integration, and relational databases.
- **Great Learning Executive Program in Artificial Intelligence and Machine Learning** (In partnership with Infosys), November 2022. Covered machine learning fundamentals, data analytics, exploratory data analysis, data preprocessing, handling missing values, and decision-making with data.